TEAM EQUINOX

Team submission report - Human-Agent Negotiation (HAN) track

Members	Roshia & Vitor Costa
Institution	Kyoto University

1. Decision Architecture

Equinox separates negotiation into three independently tuned layers:

- **Bidding policy -** a learned concession schedule fixing the self-utility band the agent will offer at the current time.
- **Opponent model -** an online estimate of the opponent's utility function that refines offers and is reported for Deception scoring.
- **Messaging -** a stance-driven layer producing human-readable chat lines without leaking private valuations.

2.1 The Bidding Policy

At the start of each negotiation, Equinox builds an observation vector describing the domain (role, step budget, outcome count, issue count, reservation value, utility spread) and feeds it to a small Gaussian policy (three 128-unit SiLU layers). The network outputs four parameters controlling the concession curve:

- **exp_min, exp_max -** two exponents defining a time-dependent utility band, covering the full Boulware/Conceder family as a continuous learnable knob.
- **floor -** a late-game utility floor defending against deadline-running hardliners.
- **delta -** a budget for the Pareto nudge - how far below the band the agent may reach for large opponent gains.

When no trained checkpoint is present the policy falls back to Boulware-style defaults. Incompatible checkpoints are detected on load and ignored with a warning.

2.2 Offer Selection and the Pareto Nudge

The concession curve fixes a utility band but not which concrete outcome to offer within it. Equinox enumerates in-band outcomes and selects the one the opponent model rates highest, breaking ties by its own utility. This is a Pareto-rational refinement: nothing is given away on the agent's side while agreement becomes more attractive to the opponent.

The nudge is gated by model confidence (trust in [0,1], derived from posterior entropy). Below a threshold the nudge is disabled; at high trust, delta lets the agent widen the band slightly downward, scaled by trust so concessions remain proportional to certainty.

2.3 Accepting and Ending

Equinox accepts an incoming offer when its normalized utility clears u_min and its raw utility exceeds the reservation value. If the best achievable outcome is below the reservation value, the agent ends the negotiation. All offering and inversion paths are wrapped so NegMAS edge cases fall back to the best known outcome rather than raising.

2. Bayesian Opponent Model

The primary model (HindriksTykhonovBayesianOpponentModel) maintains a full-joint discrete Bayesian posterior over the opponent's utility function, decomposed into three hypothesis families:

- **beta -** a grid of concession speeds capturing how quickly the opponent concedes.
- **issue-weight ranks -** all permutations of issue importance rankings.
- **evaluation curves -** per-issue shapes (monotone up, monotone down, single-peaked).

Each observed offer updates the joint posterior via a Gaussian likelihood. From the posterior, the model derives expected issue weights and evaluation curves, giving an estimated utility for any outcome - used for both the Pareto nudge and Deception scoring.

The joint posterior grows factorially with issue count. Equinox checks the projected size before building anything and falls back to NegMAS' lightweight Smith frequency model on large domains. Both models expose the same interface, so the rest of the agent is agnostic to which is in use.

3. Natural-Language Messaging

Negotiation logic and language are kept strictly separate. Each round, the agent classifies its posture into a stance (opening, holding, reciprocating, pressing, closing, walking away, or accepting) from coarse non-numeric signals. The stance - never the underlying numbers - is what gets rendered. No prices, utilities, reservation values, or issue priorities may ever reach the text.

Rendering has two tiers. The default is a bank of hand-written templates, always available. At key moments in detected live human sessions, Equinox makes a best-effort call to qwen3:4b-instruct via Ollama for more natural phrasing. That path is hardened with a short timeout, a per-negotiation call cap, a circuit breaker after the first failure, and a sanitizer that rejects any output containing digits, currency symbols, or preference-revealing tokens. Failed outputs fall back to templates; the messaging layer can never stall the negotiation.

4. Training Pipeline

The bidding policy is trained offline with Soft Actor-Critic (SAC), driven by a single YAML configuration. The reward is the real HAN score, so the policy optimizes exactly what the competition measures.

- **Scenarios -** bilateral linear-additive domains with 2-5 issues and 5-12 values per issue, generated to match the competition's generator with fixed train/validation splits.
- **Opponents -** a diverse roster spanning Boulware/Conceder/Linear/Aspiration, Naive Tit-for-Tat, MiCRO, Shochan (ANL2024), RUFL (ANL2025), and hardliners, with sampling weights tilted toward the strongest rivals.
- **Reward -** the exact HAN formula (Advantage + Deception, zero on no agreement) computed via a live NegMAS mechanism.
- **Learner -** standard SAC with twin Q-networks, replay buffer, soft target updates, entropy regularization, gradient clipping, and early stopping on a patience counter.

Training runs on CPU by design - NegMAS rollouts and the numpy opponent model dominate compute while the policy network is tiny. The best checkpoint is bundled with the submission and loaded at construction.

Acknowledgements

This work was supported by JSPS Kakenhi Grant Number JP23K28145 and JSPS Kakenhi Grant Number JP22H00533.